BIOINFORMATICS P7

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Question 1

In [1]:
```
import pandas as pd
from collections import Counter
```

In [2]:
```
seq1='RATPTRWPVGCFNRPWTKWSYDEALDGIKAAGYAWTGLLTASKPSLHHATATPEYLAALKQKSRHAA'
seq2='AAAVMMGLAAIGAAIGIGILGGKFLEGAARQPDLIPLLRTQFFIVMGLVDAIPMIAVGLGLYVMFAVA'
seq3='AADVSAAVGATGQSGMTYRLGLSWDWDKSWWQTSTGRLTGYWDAGYTYWEGGDEGAGKHSLSFAPVFVYEFAGDSIKPFIEAGIGVAAFSGTRVGDQNLGSSLNFEDRIGAGLKFANGQSVG
```

In [12]:
```
def Sum_comps(dict):
    sum = 0
    for i in dict.values():
        sum = sum + i
    return sum

def composition(seq):
    dict = {}
    for i in seq:
        if i in seq:
            dict[i] = dict.get(i, 0) + 1
        else:
            dict[i] = 1
    comp = {}
    for i in dict:
        comp[i] = (dict.get(i, 0) / Sum_comps(dict))*100
    df = pd.DataFrame(index=[i for i in comp])

#    return df.from_dict(comp)  # Return only the DataFrame
 #   return dict
    return comp
```

In [15]:
```
comp_seq1=composition(seq1)
comp_seq1
```

Out[15]:
```
{'R': 5.970149253731343,
 'A': 17.91044776119403,
 'T': 10.44776119402985,
 'P': 7.462686567164178,
 'W': 5.970149253731343,
 'V': 1.4925373134328357,
 'G': 5.970149253731343,
 'C': 1.4925373134328357,
 'F': 1.4925373134328357,
 'N': 1.4925373134328357,
 'K': 7.462686567164178,
 'S': 5.970149253731343,
 'Y': 4.477611940298507,
 'D': 2.9850746268656714,
 'E': 2.9850746268656714,
 'L': 8.955223880597014,
 'I': 1.4925373134328357,
 'H': 4.477611940298507,
 'Q': 1.4925373134328357}
```

In [16]:
```
comp_seq2=composition(seq2)
```

In [17]:
```
comp_seq3=composition(seq3)
```

In [25]:
```
final_df = pd.DataFrame(list(comp_seq1.items()), columns=['Amino_acid', 'Seq 1'])
final_df.set_index('Amino_acid', inplace=True)
final_df
```

Out[25]:

Amino_acid	Seq 1
R	5.970149
A	17.910448
T	10.447761

In [29]:
```
df_seq2 = pd.DataFrame(comp_seq2.items(), columns=['Amino_acid', 'Value2'])
df_seq3 = pd.DataFrame(comp_seq3.items(), columns=['Amino_acid', 'Value3'])

# Merge DataFrames with the existing DataFrame
final_df = pd.DataFrame(comp_seq1.items(), columns=['Amino_acid', 'Value'])
final_df = final_df.merge(df_seq2, on='Amino_acid')
final_df = final_df.merge(df_seq3, on='Amino_acid')
final_df.set_index('Amino_acid', inplace=True)
```

In [30]:
```
final_df
```

Out[30]:

Amino_acid	Value	Value2	Value3
R	5.970149	2.941176	3.311258
A	17.910448	19.117647	10.596026
T	10.447761	1.470588	4.635762
P	7.462687	4.411765	2.649007
V	1.492537	8.823529	5.298013
G	5.970149	14.705882	15.231788
F	1.492537	5.882353	5.298013
K	7.462687	1.470588	3.973510
Y	4.477612	1.470588	5.298013
D	2.985075	2.941176	5.960265
E	2.985075	1.470588	3.973510
L	8.955224	13.235294	5.960265
I	1.492537	11.764706	5.298013
Q	1.492537	2.941176	3.311258

These findings unveil several noteworthy observations:

- Each of the three sequences demonstrates a notable presence of hydrophobic residues.
- Sequence 1 is distinguished by its elevated threonine content, contributing to a somewhat hydrophilic nature.
- Sequence 1 displays a heightened proportion of basic amino acids in contrast to the other sequences.
- Sequence 3 showcases a greater concentration of acidic amino acids compared to the other sequences.
- In contrast to the other two sequences, Sequence 2 exhibits a reduced abundance of aromatic amino acids.

Question 2

```python
In [31]: amino_weights = {
    'A': 85, 'C': 115, 'D': 130, 'E': 145, 'F': 160,
    'G': 70, 'W': 200, 'H': 150, 'I': 125, 'K': 145,
    'L': 125, 'M': 143, 'N': 130, 'Y': 175, 'P': 110,
    'Q': 140, 'R': 170, 'S': 100, 'T': 115, 'V': 110
}

def molecular_weight(seq):
    mol_weight=0
    for i in seq:
        mol_weight= mol_weight+ amino_weights.get(i,0)


    return mol_weight-((len(seq)-1)*18)
```

```python
In [32]: molecular_weight(seq1)  # SEQUENCE 1

Out[32]: 7127
```

```python
In [33]: molecular_weight(seq2)  # SEQUENCE 2

Out[33]: 6529
```

```python
In [34]: molecular_weight(seq3) # SEQUENCE 3

Out[34]: 15453
```

Question 3

```python
In [126]: def comparison(seq):
    grp_a = 0
    grp_b = 0
    dict_groups = {'A': [8.47, 8.95], 'D': [5.97, 5.91], 'C': [1.39, 0.47], 'E': [6.32, 4.78], 'T': [5.79, 6.54], 'F': [3.91, 3.6
    comp_seq = composition(seq)
    for i in comp_seq:
        grp_a += abs(dict_groups[i][0] - comp_seq[i])
        #print(grp_a)
        grp_b += abs(dict_groups[i][1] - comp_seq[i])

    if grp_a <= grp_b:
        print('Sequence belongs to Group A')
    else:
        print('Sequence belongs to Group B')
    return grp_a, grp_b
```

```python
In [127]: print(comparison(seq1))

Sequence belongs to Group A
(53.634029850746266, 56.963731343283584)
```

```python
In [128]: print(comparison(seq2))

Sequence belongs to Group A
(58.94470588235294, 60.08823529411765)
```

```python
In [129]: print(comparison(seq3))

Sequence belongs to Group B
(36.942251655629136, 32.127814569536426)
```

Question 4

```python
In [194]: import pandas as pd

# List of residues
residues = ["G", "A", "V", "L", "I", "P", "F", "Y", "W", "S", "T", "C", "M", "N", "Q", "D",
            "E", "K", "R", "H"]

# Function to calculate the composition of residues in a sequence
def composition(sequence):
    # Calculate the composition of each residue in the sequence
    composition = {residue: round(100 * sequence.count(residue) / len(sequence), 2) for
                   residue in residues}
    return composition

# Function to calculate the dipeptide composition of a sequence (method 1)
def dipeptide_composition_1(sequence):
    # Initialize the counts for each dipeptide
    dipeptide_counts = {residue1: {residue2: 0 for residue2 in residues} for residue1 in
                        residues}

    # Count the occurrences of each dipeptide in the sequence
    for i in range(len(sequence) - 1):
        dipeptide = sequence[i:i + 2]
        dipeptide_counts[dipeptide[0]][dipeptide[1]] += 1

    # Calculate the composition of the sequence
    comp = composition(sequence)

    # Calculate the dipeptide composition
    dipeptide_composition = {residue1: {residue2: round(100 *
                        dipeptide_counts[residue1][residue2] / ((len(sequence) / 100) * (comp[residue1] +
                        comp[residue2])), 2) if comp[residue1] + comp[residue2] != 0 else 0 for residue2 in residues}
                        for residue1 in residues}

    # Return the dipeptide composition as a DataFrame
    return pd.DataFrame(dipeptide_composition).transpose()

# Function to calculate the dipeptide composition of a sequence (method 2)
def dipeptide_composition_2(sequence):
    # Initialize the counts for each dipeptide
    dipeptide_counts = {residue1: {residue2: 0 for residue2 in residues} for residue1 in
                        residues}

    # Count the occurrences of each dipeptide in the sequence
    for i in range(len(sequence) - 1):
        dipeptide = sequence[i:i + 2]
        dipeptide_counts[dipeptide[0]][dipeptide[1]] += 1

    # Calculate the dipeptide composition
    dipeptide_composition = {residue1: {residue2: round(100 *
                        dipeptide_counts[residue1][residue2] / (len(sequence) - 1), 2) for residue2 in residues} for
                        residue1 in residues}

    # Return the dipeptide composition as a DataFrame
    return pd.DataFrame(dipeptide_composition).transpose()

# Function to calculate the dipeptide composition of a sequence (method 3)
def dipeptide_composition_3(sequence):
    # Initialize the counts for each dipeptide
    dipeptide_counts = {residue1: {residue2: 0 for residue2 in residues} for residue1 in
                        residues}

    # Count the occurrences of each dipeptide in the sequence
    for i in range(len(sequence) - 1):
        dipeptide = sequence[i:i + 2]
        dipeptide_counts[dipeptide[0]][dipeptide[1]] += 1

    # Calculate the composition of the sequence
    comp = composition(sequence)

    # Calculate the dipeptide composition
    dipeptide_composition = {residue1: {residue2: round(100 *
                        dipeptide_counts[residue1][residue2] / ((len(sequence) * 2 / 100 * 2) * (comp[residue1] *
                        comp[residue2])), 2) if comp[residue1] * comp[residue2] != 0 else 0 for residue2 in residues}
                        for residue1 in residues}

    # Return the dipeptide composition as a DataFrame
    return pd.DataFrame(dipeptide_composition).transpose()
```

i)

Sequence 1

	G	A	V	L	I	P	F	Y	W	S	T	C	M	N	Q	D	E	K	R	H
G	0.00	0.00	0.00	10.00	20.01	0.00	0.00	14.28	0.00	0.00	0.00	20.01	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
A	6.25	12.50	0.00	11.11	0.00	0.00	0.00	0.00	6.25	6.25	15.79	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
V	20.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
L	0.00	5.55	0.00	8.33	0.00	0.00	0.00	0.00	0.00	0.00	7.69	0.00	0.0	0.00	0.00	12.49	0.00	9.09	0.00	11.11
I	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	16.68	0.00	0.00
P	0.00	0.00	16.68	0.00	0.00	0.00	0.00	0.00	11.11	11.11	8.33	0.00	0.0	0.00	0.00	0.00	14.28	0.00	0.00	0.00
F	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	50.09	0.00	0.00	0.00	0.00	0.00	0.00
Y	0.00	6.67	0.00	11.11	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	19.98	0.00	0.00	0.00	0.00
W	0.00	0.00	0.00	0.00	0.00	11.11	0.00	0.00	0.00	12.50	18.18	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
S	0.00	0.00	0.00	10.00	0.00	0.00	0.00	14.28	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	11.11	12.50	0.00
T	9.09	10.53	0.00	0.00	0.00	16.67	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	8.33	9.09	0.00
C	0.00	0.00	0.00	0.00	0.00	0.00	50.09	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
M	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	20.01	0.00
Q	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	16.68	0.00	0.00
D	16.66	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	24.96	0.00	0.00	0.00
E	0.00	7.14	0.00	0.00	0.00	0.00	0.00	19.98	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
K	0.00	5.88	0.00	0.00	0.00	10.00	0.00	0.00	11.11	11.11	0.00	0.00	0.0	0.00	16.68	0.00	0.00	0.00	0.00	0.00
R	0.00	6.25	0.00	0.00	0.00	11.11	0.00	0.00	12.50	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	14.28
H	0.00	13.33	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	16.66

Sequence 2

	G	A	V	L	I	P	F	Y	W	S	T	C	M	N	Q	D	E	K	R	H
G	5.00	8.69	0.00	21.05	11.11	0.00	0.00	0.0	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.0	9.09	0.00	0.0
A	0.00	19.23	15.79	0.00	14.29	0.00	0.00	0.0	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.0	0.00	6.67	0.0
V	6.25	5.26	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.0	0.00	0.0	27.28	0.0	0.00	12.51	0.0	0.00	0.00	0.0
L	10.52	4.54	6.67	5.55	5.88	0.00	0.00	10.0	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	10.0	0.00	9.09	0.0
I	16.67	4.76	7.15	5.88	0.00	18.19	0.00	0.0	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.0	0.00	0.00	0.0
P	0.00	0.00	0.00	8.33	0.00	0.00	0.00	0.0	0.0	0.0	0.00	0.0	12.51	0.0	0.00	20.01	0.0	0.00	0.00	0.0
F	0.00	5.88	0.00	7.69	8.34	0.00	12.51	0.0	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.0	0.00	0.00	0.0
Y	0.00	0.00	14.29	0.00	0.00	0.00	0.00	0.0	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.0	0.00	0.00	0.0
W	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.0	0.00	0.00	0.0
S	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.0	0.00	0.00	0.0
T	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.0	0.00	0.0	0.00	0.0	33.35	0.00	0.0	0.00	0.00	0.0
C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.0	0.00	0.00	0.0
M	13.33	0.00	0.00	0.00	7.70	0.00	11.12	0.0	0.0	0.0	0.00	0.0	10.00	0.0	0.00	0.00	0.0	0.00	0.00	0.0
N	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.0	0.00	0.00	0.0
Q	0.00	0.00	0.00	0.00	0.00	20.01	16.67	0.0	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.0	0.00	0.00	0.0
D	0.00	6.67	0.00	9.09	0.00	0.00	0.00	0.0	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.0	0.00	0.00	0.0
E	9.09	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.0	0.00	0.00	0.0
K	0.00	0.00	0.00	0.00	0.00	0.00	20.01	0.0	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.0	0.00	0.00	0.0
R	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.0	33.35	0.0	0.00	0.0	25.01	0.00	0.0	0.00	0.00	0.0
H	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.0	0.00	0.00	0.0

Sequence 3

	G	A	V	L	I	P	F	Y	W	S	T	C	M	N	Q	D	E	K	R	H
G	2.17	7.69	6.45	9.38	6.45	0.00	0.00	6.45	0.00	2.63	3.33	0.0	4.17	0.00	7.14	9.38	0.00	3.45	3.57	0.00
A	15.38	9.37	4.17	0.00	4.17	5.00	4.17	0.00	0.00	0.00	4.35	0.0	0.00	4.76	0.00	4.00	0.00	0.00	0.00	0.00
V	9.68	4.17	0.00	0.00	0.00	0.00	6.25	6.25	0.00	4.35	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	7.69	0.00
L	6.25	0.00	0.00	0.00	0.00	0.00	5.88	0.00	0.00	8.34	6.25	0.0	0.00	7.14	0.00	0.00	0.00	13.34	0.00	0.00
I	6.45	0.00	0.00	0.00	0.00	8.33	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	14.29	7.14	0.00	10.00
P	0.00	0.00	8.33	0.00	8.33	0.00	8.33	0.00	0.00	0.00	0.00	0.0	0.00	11.11	0.00	0.00	0.00	0.00	0.00	0.00
F	0.00	12.50	6.25	0.00	6.25	0.00	0.00	6.25	0.00	4.35	0.00	0.0	0.00	0.00	0.00	0.00	7.14	0.00	0.00	0.00
Y	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	14.29	8.70	6.66	0.0	0.00	0.00	0.00	0.00	7.14	7.14	7.69	0.00
W	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.34	0.00	0.00	0.0	0.00	0.00	9.10	20.01	8.34	0.00	0.00	0.00
S	5.26	3.23	4.35	12.50	4.35	0.00	4.35	4.35	9.53	3.33	4.55	0.0	0.00	5.00	0.00	0.00	0.00	0.00	0.00	0.00
T	10.00	0.00	0.00	0.00	0.00	0.00	0.00	13.32	0.00	4.55	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	8.33	0.00
C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
M	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12.50	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N	3.57	4.76	0.00	7.14	0.00	0.00	7.69	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	7.14	0.00	0.00	0.00	0.00
Q	0.00	0.00	0.00	0.00	0.00	11.11	0.00	0.00	0.00	10.00	8.33	0.0	0.00	10.00	0.00	0.00	0.00	0.00	0.00	0.00
D	3.13	4.00	5.88	0.00	0.00	0.00	0.00	0.00	6.67	4.17	0.00	0.0	0.00	0.00	7.14	0.00	6.67	6.67	7.14	0.00
E	6.90	4.55	0.00	0.00	0.00	0.00	7.14	0.00	0.00	4.76	0.00	0.0	0.00	0.00	0.00	6.67	0.00	0.00	0.00	0.00
K	0.00	0.00	0.00	0.00	7.14	10.00	7.14	0.00	0.00	4.76	0.00	0.0	0.00	0.00	9.10	0.00	0.00	0.00	0.00	12.52
R	0.00	4.76	7.69	14.29	7.69	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
H	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.00	0.00	5.89	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

ii)

Sequence 1

	G	A	V	L	I	P	F	Y	W	S	T	C	M	N	Q	D	E	K	R	H
G	0.00	0.00	0.00	1.52	1.52	0.00	0.00	1.52	0.00	0.00	0.00	1.52	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
A	1.52	4.55	0.00	3.03	0.00	0.00	0.00	0.00	1.52	1.52	4.55	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
V	1.52	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
L	0.00	1.52	0.00	1.52	0.00	0.00	0.00	0.00	0.00	0.00	1.52	0.00	0.0	0.00	0.00	1.52	0.00	1.52	0.00	1.52
I	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	1.52	0.00	0.00
P	0.00	0.00	1.52	0.00	0.00	0.00	0.00	0.00	1.52	1.52	1.52	0.00	0.0	0.00	0.00	0.00	1.52	0.00	0.00	0.00
F	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	1.52	0.00	0.00	0.00	0.00	0.00	0.00
Y	0.00	1.52	0.00	1.52	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	1.52	0.00	0.00	0.00	0.00
W	0.00	0.00	0.00	0.00	0.00	1.52	0.00	0.00	0.00	1.52	3.03	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
S	0.00	0.00	0.00	1.52	0.00	0.00	0.00	1.52	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	1.52	1.52	0.00
T	1.52	3.03	0.00	0.00	0.00	3.03	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	1.52	1.52	0.00
C	0.00	0.00	0.00	0.00	0.00	0.00	1.52	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
M	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	1.52	0.00
Q	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	1.52	0.00	0.00
D	1.52	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	1.52	0.00	0.00	0.00
E	0.00	1.52	0.00	0.00	0.00	0.00	0.00	1.52	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
K	0.00	1.52	0.00	0.00	0.00	1.52	0.00	0.00	1.52	1.52	0.00	0.00	0.0	0.00	1.52	0.00	0.00	0.00	0.00	0.00
R	0.00	1.52	0.00	0.00	0.00	1.52	0.00	0.00	1.52	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	1.52
H	0.00	3.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	1.52

Sequence 2

	G	A	V	L	I	P	F	Y	W	S	T	C	M	N	Q	D	E	K	R	H
G	1.49	2.99	0.00	5.97	2.99	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	1.49	0.00	0.0
A	0.00	7.46	4.48	0.00	4.48	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.00	1.49	0.0
V	1.49	1.49	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	4.48	0.0	0.00	1.49	0.00	0.00	0.00	0.0
L	2.99	1.49	1.49	1.49	1.49	0.00	0.00	1.49	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	1.49	0.00	1.49	0.0
I	4.48	1.49	1.49	1.49	0.00	2.99	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.0
P	0.00	0.00	0.00	1.49	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	1.49	0.0	0.00	1.49	0.00	0.00	0.00	0.0
F	0.00	1.49	0.00	1.49	1.49	0.00	1.49	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.0
Y	0.00	0.00	1.49	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.0
W	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.0
S	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.0
T	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	1.49	0.00	0.00	0.00	0.00	0.0
C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.0
M	2.99	0.00	0.00	0.00	1.49	0.00	1.49	0.00	0.0	0.0	0.00	0.0	1.49	0.0	0.00	0.00	0.00	0.00	0.00	0.0
N	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.0
Q	0.00	0.00	0.00	0.00	0.00	1.49	1.49	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.0
D	0.00	1.49	0.00	1.49	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.0
E	1.49	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.0
K	0.00	0.00	0.00	0.00	0.00	0.00	1.49	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.0
R	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	1.49	0.0	0.00	0.0	1.49	0.00	0.00	0.00	0.00	0.0
H	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.0

Sequence 3

	G	A	V	L	I	P	F	Y	W	S	T	C	M	N	Q	D	E	K	R	H
G	0.67	2.00	1.33	2.00	1.33	0.00	0.00	1.33	0.00	0.67	0.67	0.0	0.67	0.00	1.33	2.00	0.00	0.67	0.67	0.00
A	4.00	2.00	0.67	0.00	0.67	0.67	0.67	0.00	0.00	0.00	0.67	0.0	0.00	0.67	0.00	0.67	0.00	0.00	0.00	0.00
V	2.00	0.67	0.00	0.00	0.00	0.00	0.67	0.67	0.00	0.67	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.67	0.00
L	1.33	0.00	0.00	0.00	0.00	0.00	0.67	0.00	0.00	1.33	0.67	0.0	0.00	0.67	0.00	0.00	0.00	1.33	0.00	0.00
I	1.33	0.00	0.00	0.00	0.00	0.67	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	1.33	0.67	0.00	0.67
P	0.00	0.00	0.67	0.00	0.67	0.00	0.67	0.00	0.00	0.00	0.00	0.0	0.00	0.67	0.00	0.00	0.00	0.00	0.00	0.00
F	0.00	2.00	0.67	0.00	0.67	0.00	0.00	0.67	0.00	0.67	0.00	0.0	0.00	0.00	0.00	0.00	0.67	0.00	0.00	0.00
Y	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.33	1.33	0.67	0.0	0.00	0.00	0.00	0.00	0.67	0.67	0.67	0.00
W	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.67	0.00	0.00	0.0	0.00	0.00	0.67	2.00	0.67	0.00	0.00	0.00
S	1.33	0.67	0.67	2.00	0.67	0.00	0.67	0.67	1.33	0.67	0.67	0.0	0.00	0.67	0.00	0.00	0.00	0.00	0.00	0.00
T	2.00	0.00	0.00	0.00	0.00	0.00	0.00	1.33	0.00	0.67	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.67	0.00
C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
M	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.67	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N	0.67	0.67	0.00	0.67	0.00	0.00	0.67	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.67	0.00	0.00	0.00	0.00
Q	0.00	0.00	0.00	0.00	0.00	0.67	0.00	0.00	0.00	1.33	0.67	0.0	0.00	0.67	0.00	0.00	0.00	0.00	0.00	0.00
D	0.67	0.67	0.67	0.00	0.00	0.00	0.00	0.00	0.67	0.67	0.00	0.0	0.00	0.00	0.67	0.00	0.67	0.67	0.67	0.00
E	1.33	0.67	0.00	0.00	0.00	0.00	0.67	0.00	0.00	0.67	0.00	0.0	0.00	0.00	0.00	0.67	0.00	0.00	0.00	0.00
K	0.00	0.00	0.00	0.00	0.67	0.67	0.67	0.00	0.00	0.67	0.00	0.0	0.00	0.00	0.67	0.00	0.00	0.00	0.00	0.67
R	0.00	0.67	0.67	1.33	0.67	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
H	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.67	0.00	0.67	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

iii)

Sequence 1

	G	A	V	L	I	P	F	Y	W	S	T	C	M	N	Q	D	E	K	R	H
G	0.00	0.00	0.00	0.70	4.19	0.00	0.00	1.40	0.00	0.00	0.00	4.19	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
A	0.35	0.35	0.00	0.47	0.00	0.00	0.00	0.00	0.35	0.35	0.60	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
V	4.19	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
L	0.00	0.23	0.00	0.46	0.00	0.00	0.00	0.00	0.00	0.00	0.40	0.00	0.0	0.00	0.00	1.39	0.00	0.56	0.00	0.93
I	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	3.36	0.00	0.00
P	0.00	0.00	3.36	0.00	0.00	0.00	0.00	0.00	0.84	0.84	0.48	0.00	0.0	0.00	0.00	0.00	1.67	0.00	0.00	0.00
F	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	16.81	0.00	0.00	0.00	0.00	0.00	0.00
Y	0.00	0.47	0.00	0.93	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	2.79	0.00	0.00	0.00	0.00
W	0.00	0.00	0.00	0.00	0.00	0.84	0.00	0.00	0.00	1.05	1.20	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
S	0.00	0.00	0.00	0.70	0.00	0.00	0.00	1.40	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.84	1.05	0.00
T	0.60	0.40	0.00	0.00	0.00	0.96	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.48	0.60	0.00
C	0.00	0.00	0.00	0.00	0.00	0.00	16.81	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
M	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	4.19	0.00
Q	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	3.36	0.00	0.00
D	2.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	4.17	0.00	0.00	0.00
E	0.00	0.70	0.00	0.00	0.00	0.00	0.00	2.79	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
K	0.00	0.28	0.00	0.00	0.00	0.67	0.00	0.00	0.84	0.84	0.00	0.00	0.0	0.00	3.36	0.00	0.00	0.00	0.00	0.00
R	0.00	0.35	0.00	0.00	0.00	0.84	0.00	0.00	1.05	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	1.40
H	0.00	0.93	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	1.86

Sequence 2

	G	A	V	L	I	P	F	Y	W	S	T	C	M	N	Q	D	E	K	R	H
G	0.17	0.26	0.00	0.76	0.43	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	1.7	0.00	0.0
A	0.00	0.50	0.65	0.00	0.49	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.65	0.0
V	0.28	0.22	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	1.70	0.0	0.00	1.42	0.00	0.0	0.00	0.0
L	0.38	0.15	0.31	0.21	0.24	0.00	0.00	1.89	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	1.89	0.0	0.94	0.0
I	0.64	0.16	0.35	0.24	0.00	1.42	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.0
P	0.00	0.00	0.00	0.63	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	1.13	0.0	0.00	2.84	0.00	0.0	0.00	0.0
F	0.00	0.33	0.00	0.47	0.53	0.00	1.06	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.0
Y	0.00	0.00	2.84	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.0
W	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.0
S	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.0
T	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	8.51	0.00	0.00	0.0	0.00	0.0
C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.0
M	0.68	0.00	0.00	0.00	0.43	0.00	0.85	0.00	0.0	0.0	0.00	0.0	0.68	0.0	0.00	0.00	0.00	0.0	0.00	0.0
N	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.0
Q	0.00	0.00	0.00	0.00	0.00	2.84	2.13	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.0
D	0.00	0.65	0.00	0.94	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.0
E	1.70	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.0
K	0.00	0.00	0.00	0.00	0.00	0.00	4.25	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.0
R	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	8.51	0.0	0.00	0.0	4.25	0.00	0.00	0.0	0.00	0.0
H	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.0

Sequence 3

	G	A	V	L	I	P	F	Y	W	S	T	C	M	N	Q	D	E	K	R	H
G	0.07	0.31	0.41	0.55	0.41	0.00	0.00	0.41	0.00	0.11	0.23	0.0	1.65	0.00	0.66	0.55	0.00	0.27	0.33	0.00
A	0.62	0.44	0.29	0.00	0.29	0.59	0.29	0.00	0.00	0.00	0.34	0.0	0.00	0.47	0.00	0.26	0.00	0.00	0.00	0.00
V	0.62	0.29	0.00	0.00	0.00	0.00	0.59	0.59	0.00	0.31	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.94	0.00
L	0.36	0.00	0.00	0.00	0.00	0.00	0.52	0.00	0.00	0.56	0.60	0.0	0.00	0.84	0.00	0.00	0.00	1.40	0.00	0.00
I	0.41	0.00	0.00	0.00	0.00	1.18	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	1.57	0.79	0.00	2.37
P	0.00	0.00	1.18	0.00	1.18	0.00	1.18	0.00	0.00	0.00	0.00	0.0	0.00	1.89	0.00	0.00	0.00	0.00	0.00	0.00
F	0.00	0.88	0.59	0.00	0.59	0.00	0.00	0.59	0.00	0.31	0.00	0.0	0.00	0.00	0.00	0.00	0.79	0.00	0.00	0.00
Y	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.57	0.63	0.67	0.0	0.00	0.00	0.00	0.00	0.79	0.79	0.94	0.00
W	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.05	0.00	0.00	0.0	0.00	0.00	1.26	2.10	1.05	0.00	0.00	0.00
S	0.22	0.16	0.31	0.84	0.31	0.00	0.31	0.31	0.84	0.17	0.36	0.0	0.00	0.50	0.00	0.00	0.00	0.00	0.00	0.00
T	0.70	0.00	0.00	0.00	0.00	0.00	0.00	1.35	0.00	0.36	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	1.08	0.00
C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
M	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.41	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N	0.33	0.47	0.00	0.84	0.00	0.00	0.94	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.84	0.00	0.00	0.00	0.00
Q	0.00	0.00	0.00	0.00	0.00	1.89	0.00	0.00	0.00	1.01	1.08	0.0	0.00	1.51	0.00	0.00	0.00	0.00	0.00	0.00
D	0.18	0.26	0.52	0.00	0.00	0.00	0.00	0.00	0.70	0.28	0.00	0.0	0.00	0.00	0.84	0.00	0.70	0.70	0.84	0.00
E	0.55	0.39	0.00	0.00	0.00	0.00	0.79	0.00	0.00	0.42	0.00	0.0	0.00	0.00	0.00	0.70	0.00	0.00	0.00	0.00
K	0.00	0.00	0.00	0.00	0.79	1.57	0.79	0.00	0.00	0.42	0.00	0.0	0.00	0.00	1.26	0.00	0.00	0.00	0.00	3.16
R	0.00	0.47	0.94	1.68	0.94	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
H	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.37	0.00	1.26	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Top 10 preferred residues from each paired residues are:

In [210]:
```
print(find_top_ten(dipeptide_composition_1(seq1)))
print(find_top_ten(dipeptide_composition_1(seq2)))
print(find_top_ten(dipeptide_composition_1(seq3)))
print(find_top_ten(dipeptide_composition_2(seq1)))
print(find_top_ten(dipeptide_composition_2(seq2)))
print(find_top_ten(dipeptide_composition_2(seq3)))
print(find_top_ten(dipeptide_composition_3(seq1)))
print(find_top_ten(dipeptide_composition_3(seq2)))
print(find_top_ten(dipeptide_composition_3(seq3)))
```

```
['FN', 'CF', 'DE', 'GI', 'GC', 'VG', 'NR', 'YD', 'EY', 'WT']
['TQ', 'RT', 'VM', 'RQ', 'GL', 'PD', 'QP', 'KF', 'AA', 'IP']
['WD', 'AG', 'IE', 'YW', 'RL', 'LK', 'TY', 'KH', 'FA', 'SL']
['AA', 'AT', 'AL', 'WT', 'TA', 'TP', 'HA', 'GL', 'GI', 'GY']
['AA', 'GL', 'AV', 'AI', 'VM', 'IG', 'GA', 'GI', 'LG', 'IP']
['AG', 'GA', 'GL', 'GD', 'AA', 'VG', 'FA', 'WD', 'SL', 'TG']
['FN', 'CF', 'GI', 'GC', 'VG', 'NR', 'DE', 'IK', 'PV', 'QK']
['TQ', 'RT', 'KF', 'RQ', 'PD', 'YV', 'QP', 'QF', 'LY', 'LE']
['MT', 'KH', 'IH', 'HY', 'WD', 'PN', 'QP', 'RL', 'GM', 'IE']
```

Question 5

In [131]:

```
# Define a function to calculate the average hydrophobicity of a sequence
def hydrophobicity(sequence):
    # Define a dictionary with the hydrophobicity values for each residue
    data = {'A': 13.85, 'D': 11.61, 'C': 15.37, 'E': 11.38, 'F': 13.93, 'G': 13.34,
            'H': 13.82, 'I': 15.28, 'K': 11.58, 'L': 14.13, 'M': 13.86, 'N': 13.02,
            'P': 12.35, 'Q': 12.61, 'R': 13.10, 'S': 13.39, 'T': 12.70, 'V': 14.56,
            'W': 15.48, 'Y': 13.88}
    # Use a list comprehension to calculate the total hydrophobicity
    total = sum(data[residue] for residue in sequence)
    # Return the average hydrophobicity, rounded to 2 decimal places
    return round(total / len(sequence), 2)

# Define a function to calculate the helical contact area of a sequence
def helical_contact_area(sequence):
    # Define a dictionary with the helical contact area values for each residue
    data = {'A': 20.0, 'D': 26.0, 'C': 25.0, 'E': 33.0, 'F': 46.0, 'G': 13.0, 'H': 37.0,
            'I': 39.0, 'K': 46.0, 'L': 35.0, 'M': 43.0, 'N': 28.0, 'P': 22.0, 'Q': 36.0,
            'R': 55.0, 'S': 20.0, 'T': 28.0, 'V': 33.0, 'W': 61.0, 'Y': 46.0}
    # Use a list comprehension to calculate the total helical contact area
    total = sum(data[residue] for residue in sequence)
    # Return the total helical contact area, rounded to 2 decimal places
    return round(total, 2)

# Define a function to calculate the total non-bonded energy of a sequence
def total_nonbonded_energy(sequence):
    # Define a dictionary with the non-bonded energy values for each residue
    data = {'A': 1.9, 'D': 1.52, 'C': 2.04, 'E': 1.54, 'F': 1.86, 'G': 1.9, 'H': 1.76,
            'I': 1.95, 'K': 1.37, 'L': 1.97, 'M': 1.96, 'N': 1.56, 'P': 1.7, 'Q': 1.52,
            'R': 1.48, 'S': 1.75, 'T': 1.77, 'V': 1.98, 'W': 1.87, 'Y': 1.69}

    total = sum(data[residue] for residue in sequence)

    return round(total, 2)

properties = pd.DataFrame({seq: [hydrophobicity(seq), helical_contact_area(seq), total_nonbonded_energy(seq)] for seq in [seq1, s


properties.columns = ["Seq. 1", "Seq. 2", "Seq. 3"]

# Transpose the DataFrame
properties = properties.transpose()

# Rename the index
properties.columns = ["Average Hydrophobicity", "Helical Contact Area", "Total Non-Bonded Energy"]

# Print the DataFrame
print(properties)
```

```
        Average Hydrophobicity  Helical Contact Area  Total Non-Bonded Energy
Seq. 1                   13.35                2156.0                   117.74
Seq. 2                   13.77                2067.0                   126.66
Seq. 3                   13.42                4616.0                   267.75
```